Packages, Subjects & Domains

API collection with request payloads and response examples

Base paths: `{{baseUrl}}/api` and `{{baseUrl}}/api/v1` expose the same routes. Replace `{{baseUrl}}` with the deployed server host.
Authentication: Protected routes accept `Authorization: Bearer {{token}}`. Package read/domain endpoints use optional authentication; subject and instructor-subject routes require a valid staff JWT.
Content types: Create/update package and subject accept JSON or multipart/form-data. Other write requests use JSON.

Domain model: There is no standalone `/api/domains` router. Packages and subjects each store a text `domain` field. Package-domain endpoints list/filter domains from package records. The migration also seeds domain-like package and subject records with `type: "Domain"` (Testing, Development, Cybersecurity, Devops, AI/ML, Data Analytics, UI/UX Design).

Package APIs

GET /api/packages

Auth: Optional. **Query (optional):** `domain`, `domainInterest`, or `interest` filters packages by domain; matching is case-insensitive and supports partial text. No request body. Returns a raw JSON array with package fields and associated `subjects`.

```
[{
  "id": 12, "name": "QA Professional", "code": "QA-PRO",
  "description": "Software quality program", "type": "advance", "duration": 16,
  "image": "https://cdn.example.com/package.png", "overview": {}, "syllabus": [],
  "prerequisites": [], "startDate": "2026-11-01T00:00:00.000Z",
  "domain": "Testing", "mode": "online",
  "subjects": [{ "id": 3, "name": "Manual Testing", "code": "MT-01" }]
}]
```

GET /api/packages/domains

Auth: Optional. No body. Returns the sorted unique domain names parsed from non-null package `domain` values (comma-separated values are split).

```
["AI/ML", "Cybersecurity", "Development", "Testing"]
```

GET /api/packages/domain/:domain

Auth: Optional. **Path:** domain text, URL-encode spaces/slashes as needed. No body. Returns the same package array shape as `GET /api/packages`, filtered by domain.

```
GET /api/packages/domain/Testing
GET /api/packages/domain/Data%20Analytics
```

GET /api/packages/domain

Auth: Optional. **Query:** `domain` (or alias `domainInterest`/`interest`) required. No body.

```
GET /api/packages/domain?domain=Testing
```

Returns matching packages as a raw JSON array; missing domain returns `400 { "message": "Domain parameter is required" }`.

GET /api/packages/:id

Auth: Optional. **Path:** package ID. No body. Returns one package with associated subjects; missing package returns 404.

```
{ "id": 12, "name": "QA Professional", "code": "QA-PRO", "domain": "Testing", "subjects": [{ "id": 3, "name": "Manual Testing", "code": "MT-01" }] }
```

GET /api/packages/:id/price

Auth: Required. **Path:** package ID. No body. Despite the route name, the current handler returns package summary fields; the model/controller does not include a price field.

```
{ "id": 12, "name": "QA Professional", "code": "QA-PRO", "description": "Software quality program", "type": "advance", "duration": 16, "packageType": "standard" }
```

POST /api/packages

Auth: Required; only ADMIN and COUNSELLOR may create. **Body:** JSON or multipart/form-data. Required: `name`, `code`. Optional fields: `description`, `type` (starter|advance|expert), numeric `duration`, `startDate`, `packageType` (standard|others, default standard), `overview`, `syllabus`, `prerequisites`, `domain`, `mode`, `subjectIds` (array of existing subject IDs), and `image` file (multipart only, max 10MB).

```
{
  "name": "QA Professional", "code": "QA-PRO",
  "description": "Software quality program", "type": "advance",
  "duration": 16, "startDate": "2026-11-01",
  "packageType": "standard", "domain": "Testing", "mode": "online",
  "overview": { "summary": "Testing fundamentals to automation" },
  "syllabus": ["Test design", "Automation"],
  "prerequisites": ["Basic computer skills"], "subjectIds": [3, 4]
}
```

For `packageType: "others"`, `subjectIds` must be a non-empty array. Multipart `subjectIds` may be a JSON array string or comma-separated IDs.

```
{ "message": "Package created successfully", "package": { "id": 12, "name": "QA Professional", "code": "QA-PRO", "subjects": [{ "id": 3, "name": "Manual Testing", "code": "MT-01" }] } }
```

PUT /api/packages/:id

Auth: Required; only ADMIN and COUNSELLOR may update. **Path:** package ID. **Body:** JSON or multipart/form-data; all fields are optional and follow POST names/types. `subjectIds`, when sent, replaces the package's subject links and must be a non-empty array of existing IDs. Changing `packageType` to `others` requires subjectIds. Image upload replaces the existing image.

```
{ "domain": "Development", "mode": "hybrid", "subjectIds": [8, 9] }
```

Response (200): `{ "message": "Package updated successfully", "package": { ...updatedPackage, "subjects": [...] } }`.

DELETE /api/packages/:id

Auth: Required; only ADMIN and COUNSELLOR may delete. No body. Clears enquiry references and linked subjects, then deletes the package.

```
{ "message": "Package deleted successfully" }
```

Subject APIs

GET `/api/subjects`

Auth: Required. No body. Returns a raw JSON array of subjects with fields including `domain`, `mode`, and associated `packages`.

GET `/api/subjects/:id`

Auth: Required. **Path:** subject ID. No body. Returns one subject including associated package summaries; 404 if missing.

GET `/api/subjects/subjectby-instructor`

Auth: Required. Uses instructor ID from JWT; no body. Returns `{ success, message, total, data[] }` with each subject's package summaries. If no assignments: `{ success: true, message: "No subjects found for this instructor", data: [] }`.

GET `/api/subjects/fees/byids`

Auth: Required. IDs may be sent as a comma-separated or repeated query parameter. Although handler code also reads `req.body.ids`, prefer query parameters for GET.

```
GET /api/subjects/fees/byids?ids=3,4
GET /api/subjects/fees/byids?ids=3&ids=4
```

Alternative body supported by handler: `{ "ids": [3, 4] }`. Response is `{ "count": 2, "data": [{ "id": 3, "name": "Manual Testing", "code": "MT-01", "description": "...", "type": "starter", "duration": 4 }] }`. The current Subject model has no fee/price field, so the endpoint returns identifying/description/type/duration fields rather than prices.

POST `/api/subjects`

Auth: Required; only ADMIN and COUNSELLOR may create. **Body:** JSON or multipart/form-data. Required: `name`, `code`. Optional: `description`, `type` (starter|advance|expert), numeric `duration`, `startDate`, `overview`, `syllabus`, `prerequisites`, `domain`, `mode`, and multipart `image` (max 10MB).

```
{ "name": "Manual Testing", "code": "MT-01", "description": "Testing foundations", "type": "starter",
"duration": 4, "startDate": "2026-11-01", "domain": "Testing", "mode": "online", "overview": { "summary":
"Core QA concepts" }, "syllabus": ["Test cases", "Defects"], "prerequisites": [] }
```

Response (201): `{ "message": "Subject created successfully", "subject": { ...createdSubject } }`.

PUT `/api/subjects/:id`

Auth: Required; only ADMIN and COUNSELLOR may update. **Path:** subject ID. Same fields as create, all optional; JSON or multipart/form-data. Supplied image replaces the prior image.

```
{ "domain": "Testing", "mode": "hybrid", "duration": 6 }
```

Response (200): `{ "message": "Subject updated successfully", "subject": { ...updatedSubject } }`.

DELETE `/api/subjects/:id`

Auth: Required; only ADMIN and COUNSELLOR may delete. No body. Removes subject references from enquiry subject ID arrays, then deletes subject.

```
{ "message": "Subject deleted successfully" }
```

Instructor-Subject Association APIs

POST `/api/instructor-subjects`

Auth: Required; only ADMIN and COUNSELLOR may assign. JSON body:

```
{ "instructorId": 12, "subjectId": 3 }
```

Both IDs required and both records must exist. Duplicate mapping returns 400.

```
{ "success": true, "message": "Subject assigned to instructor successfully", "data": { "id": 2,
"instructorId": 12, "subjectId": 3 } }
```

GET `/api/instructor-subjects`

Auth: Required. No body. Response: `{ success, message, totalMappings, data[] }`; mappings include instructor and subject summaries.

GET `/api/instructor-subjects/instructor/:instructorId`

Auth: Required. Path instructor ID; no body. Returns `{ success, message, instructor, subjectCount, data[] }`.

GET `/api/instructor-subjects/subject/:subjectId`

Auth: Required. Path subject ID; no body. Returns `{ success, message, subject, instructorCount, data[] }`.

DELETE `/api/instructor-subjects/:id`

Auth: Required; only ADMIN and COUNSELLOR may remove assignments. Path mapping ID; no body. Response: `{ "success": true, "message": "Subject removed from instructor successfully" }`.

Common Validation and Errors

Status	Typical cause
400	Missing required name/code/domain/IDs, invalid type/duration/packageType, duplicate code/mapping, empty subjectIds, or invalid request data.
403	Authenticated role is not permitted to create/update/delete/assign.
404	Package, subject, instructor, or one of the selected subjects was not found.
500	Unexpected server/database/upload error.

This reference reflects current routes/controllers/models. “Price” endpoints and fee lookup names are documented as implemented; current package/subject models do not define an actual price/fee property. Domain names listed above come from a seed migration and may differ in databases where that migration was not run.